

MANTEIS SYSTEMS

2026 EDITION

How to deploy production-grade AI entirely on-premises

zero bytes to the cloud, full regulatory compliance, lower lifetime cost.

FOR REGULATED INDUSTRIES / LAW / HEALTHCARE / FINANCE / ENGINEERING

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EXECUTIVE SUMMARY

This guide compresses two decades of enterprise IT, security auditing, and systems engineering into a documented, deployable architecture for running production AI entirely on hardware you own. It is not a product pitch. It is a blueprint you can evaluate, take to your CIO, and use to make a build-vs-buy decision.

SECTION 01

The Sovereignty Problem

Why cloud AI fails the buyers who need it most

There is a quiet bifurcation happening in enterprise AI. On one side: the cloud AI vendors, racing to commoditize inference, dropping per-token prices, and offering integration that means one thing: send us your data and we will send back answers.

On the other side: a growing cohort of businesses that will never, under any circumstances, send their data to a third party. Not because they are paranoid. Because their regulators, their clients, and their own risk officers will not allow it.

These are law firms handling privileged communications. Healthcare clinics bound by HIPAA. Financial institutions with fiduciary obligations. Manufacturers protecting proprietary processes. Religious organizations safeguarding member data.

The vendor response has been inadequate.

Cloud AI vendors offer private endpoints and data residency, but the data still leaves your network. The technology to run production-grade AI entirely on-premises exists today. It is mature, open-source, and deployed in real businesses. This guide documents it.

SECTION 02

The Local-First Stack

Six open-source components that compose into production AI infrastructure

No vendor lock-in. No per-seat subscriptions. No data leaving your network. Here are the six components:

HERMES (NOUS RESEARCH)

Agent Runtime

The autonomous intelligence layer. Runs scheduled operations, triages, and multi-step workflows. The active operator that makes the system think, not a chatbot, but an agent that can monitor, triage, and execute through a human-gated approval framework.

OLLAMA

Model Serving

Serves large language models locally: the same model families the cloud vendors offer (GLM, Kimi, DeepSeek, Llama, Mistral), running on hardware you own, inside your network, with zero bytes transmitted externally.

CHROMADB / QDRANT

Vector Storage

Persistent semantic search across your documents, SOPs, and knowledge base. Enables RAG pipelines and agent memory that compound over time.

N8N

Workflow Orchestration

Visual workflow engine for automated document processing, email classification, intelligent routing, and approval chains. 5+ pre-built workflows out of the box.

DOCKER

Containerization

Full containerized stack: MCP servers, vector databases, monitoring agents, custom services. Reproducible across macOS, Linux, and Windows.

MCP

Tool Protocol

Model Context Protocol: the standardized interface connecting AI models to your existing systems: Jira, M365, Active Directory, SQL databases, ERP systems, and any custom tool.

SECTION 03

Architecture Blueprint

A production reference: no confidentiality required

The architecture below runs in production today. Every component is open-source and deployable by any qualified engineering team. All AI compute and data storage stays inside the firm network boundary.

FIRM INTERNAL NETWORK

OLLAMA (LLM) + CHROMADB (VECTORS) + N8N (ORCH) + MCP SERVERS

HERMES AGENT (AUTONOMOUS OPS: READ-ONLY + GATED WRITES)

TOOL INTEGRATIONS: M365/AD + JIRA/TICKETS + SQL/ERP (GUARDED) + SIEM

SOVEREIGN OS WEB INTERFACE (sovereign.local): DASHBOARD / CHAT / WORKFLOWS / SECURITY

FIREWALL (NGFW + IDS/IPS): BLOCKS ALL AI API EGRESS

Key principle: Hermes handles reasoning and orchestration. Ollama handles inference on local hardware. Data classification determines which workloads run on which layer. Every tool integration is read-only by construction: writes go through a human-gated approval queue with full audit trail.

SECTION 04

Security: The Fortress Protocol

Zero-trust before deployment

Network Segmentation

AI infrastructure runs on an isolated VLAN. Outbound rules block all AI API endpoints. Only DNS, package repos, and approved vendor updates are permitted. Inbound access via zero-trust mesh (Tailscale) with SSO.

Read-Only by Construction

Every tool integration is built read-only. SQL queries pass through a guard that strips comments, rejects non-SELECT statements, and blocks 25+ forbidden keywords. The agent cannot write to any system without human approval.

Human-Gated Writes

The agent proposes actions. A human approves them. A closed registry of executors runs the approved action with full JSON audit trail. Low-risk actions can be auto-approved through a double-gate; production actions always require human approval.

Endpoint Security

All endpoints run EDR. The platform monitors sign-in failures with password-spray and brute-force heuristics. Mailbox forwarding rules are audited for BEC indicators. Defender alerts flow into the SIEM automatically.

Data Classification

Workloads are routed by data classification tier. Public/internal data can use cloud APIs. Confidential/restricted data must run on local infrastructure. Enforced in code, not documentation.

Full Disk Encryption

Every node uses FDE. Backups are encrypted. NAS uses RAID. UPS battery backup ensures the AI does not die when the power blinks.

SECTION 05

Cost Analysis: Sovereign vs Cloud

The numbers over 3 years

Cloud AI is a subscription: you pay forever, the price never goes down, and you own nothing. Sovereign AI is an investment: you pay once, the infrastructure is yours, and the system compounds in capability without compounding in cost.

	Cloud SaaS	Sovereign AI	
Year 1	\$144,000	\$85,000	Hardware + deployment
Year 2	\$144,000	\$0	You own it
Year 3	\$144,000	\$0	Or optional retainer
3-Year Total	\$432,000	\$85,000	84% savings
You own	Nothing	Everything	

Year one is cost-comparable. Year two and beyond, you are dramatically cheaper: because you are not paying per-seat for inference that runs on hardware you already own.

The firm that owns its AI infrastructure has a competitive moat the firm paying a monthly SaaS subscription does not.

SECTION 06

Deployment Roadmap

A phased implementation plan

Every deployment is structured as fixed-scope phases with clear deliverables and exit criteria. You decide after each phase whether to continue.

PHASE 1: Business Process Automation (4-8 weeks)

Local LLM deployment, 5+ automated workflows, document automation, intelligent routing. Exit: 50-70% reduction in manual processing.

PHASE 2: Autonomous IT Operations (6-12 weeks)

AI agent integrated with ticketing, real-time triage, technician guidance, MDM fleet integration, scripted fix delivery with human approval. Exit: 50%+ tier-1 tickets auto-resolved.

PHASE 3: Autonomous Security Operations (8-16 weeks)

Full SIEM, AI-powered threat triage, automated containment, endpoint agents, pfSense integration. Exit: SOC without hiring a SOC team.

PHASE 4: Proactive Infrastructure (4-8 weeks)

Predictive monitoring, self-healing, infrastructure health monitoring. Exit: 70%+ fewer reactive tickets.

PHASE 5: AI-Powered ERP Tooling (8-16 weeks)

Natural language interface on legacy ERP, intelligent data entry, custom web frontends, automated reporting.

Hardware right-sizing: 10 tiers from compact mini-PC (Best Buy, same-day) to quad-GPU enterprise server. Standard, warrantied gear. Configured and branded as a Manteis Custom Config. macOS, Linux, or Windows: your shop, your choice.

SECTION 07

The Manteis Engagement Model

Audit / Deploy / Manage

How it starts

- 1. AI Readiness Assessment (FREE):** 30-minute call. We map your use cases, regulatory constraints, and infrastructure against the local-first stack.
- 2. Phase Proposal:** Fixed scope, fixed timeline. No open-ended consulting contracts.
- 3. Phase-by-Phase Progression:** Each phase has deliverables, exit criteria, and 30-day support. Knowledge transfer is built in: your team owns the system when we leave.

What included in every phase

Full deployment documentation

Staff training and knowledge transfer

Source code and configuration files

30 days post-deployment support

Your team owns the system when we leave

The optional retainer

Monthly retainer for ongoing optimization, monitoring, and workflow extension. Not required. The system runs without us. But most clients want the engine to keep improving.

No pricing on the website. Every engagement is scoped to your budget, timeline, and requirements. From small business to enterprise: the architecture scales, the cost flexes, the sovereignty stays absolute.

SECTION 08

Decision Framework

Should you go sovereign?

Sovereign AI is not for everyone. But if any of the following are true, it is worth evaluating:

Your data cannot leave your network (regulatory, contractual, or fiduciary obligation)

You are paying more than 50K/year in cloud AI subscriptions

Your team needs AI integrated with internal systems (ERP, AD, ticketing, file shares)

You want AI that compounds in capability without compounding in cost

You have been told we cannot use AI because of compliance and you do not accept that answer

You want to own your AI infrastructure the way you own your servers

The next step

Book a discovery call. 60 minutes. No pitch deck. We audit your current infrastructure, identify where overhead is bleeding your time, and propose the right first step.

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**Your intelligence should be an asset, not a
subscription.**